Testing the Tau-Core Observer Principle with Gravitational Lensing: Theoretical Feasibility, Data Audit, and Missing Evidence-Grade Products

Draft date: 2026-05-21
Repository branch: `codex/tau-core-lensing-observer-gate`
Status: Paper 1-style manuscript draft; theoretical test criterion established; real-data proof not yet claimed

Abstract

The tau-core observer principle is treated here as an operational working hypothesis, not as a completed covariant field theory. In standard lensing language, the paper studies a restricted perturbation of the image-level Fermat surface: an effective latent operator whose coefficient is shared across image systems rather than absorbed into arbitrary per-lens nuisance freedom. The working hypothesis is that a gravitational measurement is not only a property of the source-side mass distribution: it is also a statement about which projection channels are accessible to the observer. Strong gravitational lensing is a natural place to test this principle because a single background source can be observed through several images, each with its own parity, Fermat potential, path geometry, and measured time delay. In this paper we ask a deliberately limited question: is it theoretically possible to turn the tau-core observer principle into a falsifiable gravitational-lensing test, what data have we used to stress that test, and which data products are still missing? The proposed test observable is a shared amplitude, `epsilon_tau`, multiplying the channel `T2_parity_abs_fermat`: a parity-weighted, absolute-Fermat term that must be common across lenses and must remain distinguishable from ordinary lens-model nuisance freedom.

We make three contributions. First, we define the necessary proof gate: a no-T2 baseline must reproduce a published time-delay distance or null posterior from evidence-grade image/model products before any tau-core perturbation is allowed. Second, we show that the proposed T2 channel is mathematically identifiable in controlled synthetic lensing experiments. In a 28-lens hierarchical run with 224 position-constraint rows, an injected `epsilon_tau=0.05` is recovered as `epsilon_hat=0.0499193089`; blind and cached stress tests retain the active T2 branch under core nuisance perturbations. Third, we close a static-lens control branch using HFF cluster morphology controls. These controls do not prove T2, but they show that the workflow does not automatically manufacture tau-like signals in systems without a time-delay posterior.

The present paper does not claim that the observer principle has been proven in real lensing data. Instead, it establishes the theoretical test pathway and the current data gap. A no-contact public-product audit found no target with

enough public image/model product depth to support the required no-T2 reproduction gate. HE0435-1223 public imaging can be ingested, but the public-only PSF/model route fails validation; additional probes of RXJ1131-1231, WFI2033-4723, WGD2038-4008, and DES J0408-5354 remain product-blocked. The result is therefore a feasibility-and-reproducibility paper: the lensing test is well-defined and stress-tested, but the evidence-grade public products needed for a real-data proof are still missing.

Author and AI-Assistance Statement

This manuscript is an auditable methodological draft. The originating author is an independent researcher and does not claim institutional expertise in gravitational lensing, cosmography, or modified gravity. The hypothesis, test priorities, and research direction originated from the author and were developed with AI assistance into scripts, tables, summaries, and prose. AI assistance is not peer review or expert validation. The work should be evaluated by reproducibility, conservative claim boundaries, source traceability, and independent expert audit.

1. Introduction: The Tau-Core Observer Principle

The tau-core observer principle begins from a simple measurement distinction. In a gravitational system there is a physical source, such as a galaxy, cluster, or lensing mass distribution. But an observer never measures the source in itself. The observer measures light paths, arrival times, image positions, frequency shifts, projected shapes, and other accessible channels. The tau-core idea is that some gravitational residuals may live not in a hidden mass component alone, and not in a source-only modification alone, but in the mapping between source geometry and observer-accessible projections.

In this language, `tau` is not introduced as an extra directly observed clock that can simply be read off from data. In the present paper, `tau` is treated as an operational latent projection coordinate: no claim is made here that it is a covariant field, a gauge-invariant degree of freedom, or a completed dynamical sector. The operational question is therefore not "can we see tau directly?" but:

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Can we find an observer-accessible channel whose geometry requires a shared
tau-like projection amplitude, after ordinary source/lens nuisance freedom has
been controlled?
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This is why gravitational lensing is attractive. A lens does not give one measurement of one source. It gives multiple images of the same source through different light paths. In a time-delay lens these paths have different arrival times and different Fermat-potential structure. In a quad lens they also have a rich image-parity structure. Thus, a lens is almost a natural interferometer for observer-projection ideas: several observed channels constrain one underlying source-lens configuration.

The tau-core observer principle would become physically meaningful in lensing only if it predicts a constrained cross-image or cross-lens structure. If every lens were allowed to choose its own arbitrary correction, the idea would be empty. The test must therefore look for a shared amplitude, common across lenses or image systems, attached to a geometrically defined channel.

In this paper the candidate channel is:

`T2_parity_abs_fermat`

It combines three ingredients that are natural in time-delay lensing:

1. image parity;
2. Fermat-potential differences;
3. relative time-delay structure.

The associated amplitude is:

`epsilon_tau`

The goal of the present paper is not to claim that `epsilon_tau` has been measured. The goal is narrower and, at this stage, more defensible:

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1. formulate the lensing test implied by the observer principle;
2. show that the test is identifiable in synthetic lensing systems;
3. show that static controls do not automatically generate false positives;
4. audit which public data products were actually tested;
5. state precisely which evidence-grade products are still missing.
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With that framing, the paper is a theoretical feasibility and data-readiness study for a future proof, not a proof claim itself.

2. Toward a Physical Theory: Minimal Effective Formulation

The largest weakness of the present theory is also the reason this paper is framed as a proof-gate manuscript rather than as a modified-gravity discovery paper. The tau-core observer principle is not yet supplied with a complete physical definition. However, the idea can be made more precise than a philosophical statement. This section separates three levels. Level 0 is the measurement principle; Level 1 is the phenomenological tau-lensing model used in this paper; Level 2 is a complete covariant tau-core gravitational theory.

The present paper works at Level 1. It uses Level 0 as motivation and treats Level 2 as an open theory-completion problem.

2.1 Measurement Principle

The measurement principle is that an observer does not access the gravitational source directly. The observer accesses a finite set of projection channels gen-

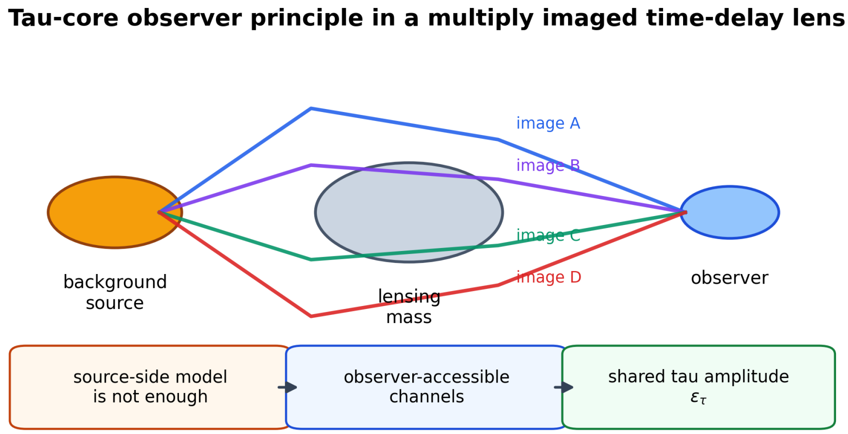

Figure 1: Figure 1. Schematic of the tau-core observer principle in a multiply imaged time-delay lens. The same background source reaches the observer through several image channels, making parity, Fermat structure, and time delays observer-accessible probes rather than source-only quantities.

erated by null geodesics, arrival-time surfaces, image parities, redshifts, and detector conventions.

The tau-core claim is that some residual structure may be attached to this observer-accessible projection map rather than to a new source-side mass component alone. In other words, a successful model must not merely change the mass distribution; it must predict a constrained correction to the map between the source-lens geometry and the observed channels.

This principle is still broad. It becomes physical only after we specify what object carries the correction, how it transforms, how it couples to light or to the lensing potential, and how the GR limit is recovered.

2.2 Relational Observable Ansatz

The minimal phenomenological ansatz used here is not a full field theory, but it identifies the object that a future field theory would have to reproduce. We introduce a latent relational projection variable:

$$\tau(x; u).$$

The notation is deliberately conservative. The paper does not decide whether `tau` is a scalar field, foliation variable, hidden clock, boundary degree of freedom, information variable, or gauge artifact. It treats `tau(x; u)` as a relational observable ansatz: `x` labels the spacetime point or lens-plane location used by the model, while `u` labels the convention by which an observer-accessible projection is constructed.

The symbol `u` should not be read as a new preferred cosmic frame. It is a bookkeeping label for a gauge-fixed observer construction, analogous in spirit to choosing a tetrad, screen basis, source-plane convention, reference image, or detector frame before reporting lensing observables. A completed covariant theory would have to show how this construction transforms when the observer description is changed and which combinations are invariant. The present paper uses `u` only to mark that the proposed residual is relational rather than source-only.

This avoids the strongest premature interpretation. We are not postulating a Lorentz-violating aether field, a universal foliation, or a directly observable hidden clock. Those possibilities are not ruled out by the notation, but they are not claimed here. The only object used in the lensing test is the first-order projection of this relational structure onto image-level Fermat and parity data.

At the effective lensing level, the paper does not require the full function `tau(x; u)`. It requires only the first-order image-channel projection:

$$\delta\Phi_i = \epsilon_\tau T_i^{(2)}, \qquad T_i^{(2)} = \pi_i |\Delta\phi_i|.$$

Here `Phi_i` is the effective arrival-time/Fermat structure for image `i`, `Delta phi_i` is the model-derived Fermat-potential difference relative to a reference image, and `\pi_i` is the image parity or parity proxy. This is the bridge between the broad observer principle and the actual data test.

2.3 Tau-Core Projection Kernel

The older tau-core/TPG construction treated the observable correction as a projection-window integral over a hidden or relational time coordinate, not as a directly visible matter component. In its simplest form, a projected observable correction has the structure

$$\mathcal{P}[X] = \int_0^{T[X]} w(\tau)\, d\tau,$$

where X is the ordinary observable that sets the depth of the projection window. The tau-core symmetry assumption is dilation covariance on positive τ-space:

$$w(\lambda\tau)\, d(\lambda\tau) = w(\tau)\, d\tau.$$

Up to regularisation at the origin, this selects the scale-invariant density $w(\tau) \propto 1/\tau$. The regularised minimal representative is

$$w(\tau) = \frac{\alpha}{1+\tau}.$$

Therefore a dimensionless projection depth T produces

$$\mathcal{P}(T) = \alpha \int_0^T \frac{d\tau}{1+\tau} = \alpha \ln(1+T).$$

For lensing, the natural image-level candidate for a dimensionless projection depth is the Fermat offset normalized by a lens time-delay scale:

$$T_i = \frac{|\Delta\phi_i|}{\phi_\star}.$$

In the small-window limit $T_i \ll 1$,

$$\mathcal{P}(T_i) = \alpha T_i + \mathcal{O}(T_i^2) = \frac{\alpha}{\phi_\star}|\Delta\phi_i| + \mathcal{O}(|\Delta\phi_i|^2).$$

This supplies the tau-core origin of the absolute Fermat factor: it is the leading term of a dilation-covariant projection-window integral when the projection depth is the image's Fermat offset. The remaining sign/orientation information cannot come from T_i, because T_i is a magnitude. The available local orientation invariant of the arrival-time surface is the Morse parity $\pi_i = \operatorname{sgn}\det H_i$. Combining the tau projection magnitude with Morse orientation gives

$$\delta\Phi_i = \epsilon_\tau\, \pi_i |\Delta\phi_i| + \mathcal{O}(\epsilon_\tau |\Delta\phi_i|^2, \epsilon_\tau^2).$$

This is still not a complete derivation from a covariant action. It is, however, a more specific tau-core physical route: dilation covariance supplies the projection kernel, the Fermat offset supplies the lensing projection depth, and Morse parity supplies the local orientation sign.

2.4 Effective Action Template

A complete theory would require an action. This paper does not claim to have one. The minimal action template that would be compatible with the present lensing test has the schematic form:

$$S = S_{\mathrm{GR}}[g] + S_{\mathrm{matter}}[g,\psi] + S_\tau[g,\tau,u] + \epsilon_\tau S_{\mathrm{coupling}}[g,\tau,u,k].$$

where `g` is the metric, `psi` denotes ordinary matter fields, `u` is the gauge-fixed observer/projection construction, and `k` is the photon/null direction. The coupling term must reduce, in the weak-field thin-lens limit, to an image-channel correction proportional to `parity_i * |Delta phi_i|`.

This template does not define the dynamics. It defines a consistency target: any future tau-core action that wants to explain the lensing T2 channel must produce the same first-order effective observable, while also satisfying covariance, conservation, and the GR limit.

2.5 Arrival-Time Functional Route

A more concrete variational route can be stated at the lensing-functional level. In ordinary time-delay lensing, the image branches are stationary points of an arrival-time functional. The tau-core correction can be represented phenomenologically as an additional branch functional:

$$\mathcal{A}_{\mathrm{eff}}[\theta] = \mathcal{A}_{\mathrm{GR}}[\theta] + \epsilon_\tau \mathcal{A}_\tau[\theta; u].$$

For the weak-field thin-lens reduction, the GR part is proportional to the Fermat potential,

$$\mathcal{A}_{\mathrm{GR},i} \propto \phi_i.$$

The tau-core ansatz is that the correction does not add a new projected mass component directly. Instead, it weights the observer-accessible branch by the tau projection kernel and by the local Morse orientation of the stationary point:

$$\mathcal{A}_{\tau,i} = \pi_i\, \phi_\star\, \mathcal{P}\left(\frac{|\Delta\phi_i|}{\phi_\star}\right).$$

Using the dilation-covariant kernel from the previous subsection gives

$$\mathcal{A}_{\tau,i} = \pi_i\, \phi_\star\, \alpha \ln\left(1 + \frac{|\Delta\phi_i|}{\phi_\star}\right).$$

For $|\Delta\phi_i|/\phi_\star \ll 1$,

$$\mathcal{A}_{\tau,i} = \alpha\, \pi_i |\Delta\phi_i| + \mathcal{O}\left(\pi_i \frac{|\Delta\phi_i|^2}{\phi_\star}\right).$$

Absorbing the dimensionless normalization into ϵ_τ yields the T2 perturbation. This gives a concrete variational interpretation of the operator: T2 is the leading branch contribution of a tau-projection term added to the arrival-time functional.

The stationarity condition becomes

$$\nabla_\theta \left(\phi + \epsilon_\tau \pi\, \phi_\star \mathcal{P}\left(\frac{|\Delta\phi|}{\phi_\star}\right)\right) = 0.$$

In the present paper we do not solve this modified lens equation. The operational test uses the first-order correction evaluated on the baseline GR/lens-model image branches. This is equivalent to a perturbative treatment:

$$\theta_i = \theta_i^{(0)} + \epsilon_\tau \theta_i^{(1)} + \mathcal{O}(\epsilon_\tau^2),$$

while the current T2 design keeps only the branch-level shift in the arrival-time/Fermat observable. A full theory would need to propagate this correction consistently into image positions, magnifications, time delays, and posterior lens parameters.

2.6 Symmetry, Covariance, and Conservation Requirements

A physically acceptable Level-2 theory should satisfy at least the following requirements:

Requirement	Minimal condition
diffeomorphism covariance	the full action must be coordinate independent
observer-construction covariance	changes of gauge-fixed observer/projection convention must have a defined transformation law
GR limit	all tau corrections vanish or become unobservable as `epsilon_tau -> 0`
conservation	modified field equations must imply a consistent stress-energy balance
lensing consistency	null propagation must remain well-defined and causal
no arbitrary per-lens freedom	the tau amplitude must be shared or hierarchically constrained

In the present paper, only the last two operational requirements are directly tested. The others are theory-completion gates.

2.7 Gauge Meaning

The gauge problem is central. Fermat potentials, image labels, and parity assignments are not raw detector counts; they are products of a lens model. Therefore `T2_i` cannot be treated as gauge-invariant merely because it is computable. A defensible empirical claim would require fixed image-label and reference-image conventions, a reproducible lens-model posterior, parity derived from the same model family, Fermat differences propagated through posterior uncertainty, and stability of the T2 result under allowed reparameterizations.

Until this is done, the tau-lensing model is best read as a gauge-controlled phenomenological perturbation, not as a fundamental observable.

2.8 What Is Still Open

The Level-2 theory remains incomplete. In particular, this paper does not yet provide:

Required theory ingredient	Present status
covariant action	template only
field content	relational projection ansatz `tau(x; u)` only
covariance structure	required but not derived
defining symmetry	required but not derived
conservation relation or Ward identity	required but not derived
derivation of the GR limit	not derived; imposed operationally by `epsilon_tau -> 0`
gauge meaning of `tau`	partially specified as a model-convention stability requirement

This is a hard claim boundary. At present, the protocol is intentionally stronger than the theory. This is a feature of the paper's claim boundary: it turns an under-specified principle into a falsifiable data requirement instead of promoting it as a finished theory.

The minimal object defined in this paper is therefore not a new action. It is an operational perturbation test on top of an ordinary lensing baseline. The baseline lens model supplies image parity, Fermat structure, and time-delay constraints; the observer-channel design vector is `T2_parity_abs_fermat`; the shared scalar amplitude is `epsilon_tau`; and the GR/null limit is recovered when $\epsilon_\tau = 0$.

In this operational version, `T2_parity_abs_fermat` is a predefined design vector computed from lens-model products, and `epsilon_tau` is a shared scalar coefficient. The test asks whether a single shared coefficient can account for structured residuals after a no-T2 baseline and nuisance freedom have already been controlled. This is not yet a derivation from first principles. It is a falsifiable bridge between the observer principle and data.

The paper's theory-status statement is therefore: a complete tau-core field theory is not yet supplied; the tau-lensing perturbation test is defined only at phenomenological Level 1; and real-data evidence is unavailable until a no-T2 reproduction is achieved.

3. Bridge to Standard Lensing and Inference Language

The terminology used above is intentionally internal to the tau-core program, but the statistical object is familiar. The proposed test can be read in four standard languages.

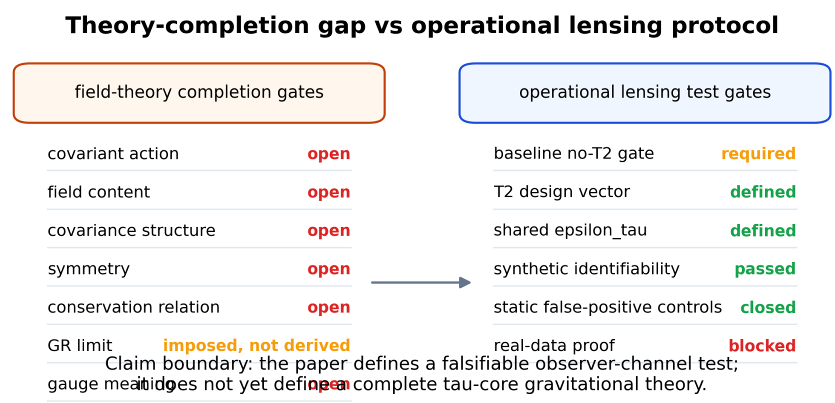

Figure 2: Figure 2. Theory-completion gap. The operational observer-gated lensing protocol is defined, while the field-theory ingredients needed for a complete physical tau-core model remain open gates.

First, in lensing perturbation language, the model is a constrained perturbation of the Fermat surface. A conventional time-delay lensing analysis models image positions, magnifications, and delays through a lens potential and a Fermat potential. The tau-lensing term does not replace that baseline. It asks whether a specific first-order residual term, tied to parity and Fermat geometry, is favored after the baseline model has already reproduced the published time-delay-distance or null posterior:

$$\Phi_i = \Phi_i^{\mathrm{GR+lens}} + \epsilon_\tau O_i + N_i, \qquad O_i = \pi_i |\Delta\phi_i|.$$

Second, in effective-operator language, `T2_parity_abs_fermat` is an empirical operator on image-level lens-model products. The operator is not arbitrary: it uses quantities already present in time-delay lensing, and it is tested against nearby controls that should not be promoted if the effect is merely absorbing modeling freedom. However, it is not yet theoretically inevitable. A complete dynamical theory would need to explain why parity, absolute Fermat structure, and a shared scalar coefficient are selected rather than another operator in the same image-level family.

Third, in Bayesian hierarchical language, `epsilon_tau` is a population-level coefficient. Individual lenses may have their own ordinary nuisance parameters, but the tau coefficient is not allowed to float independently for every lens or image. This is the key statistical distinction between a physical shared effect and a flexible residual-fitting term.

$$d_l \sim p(d_l \mid \theta_l, \eta_l, \epsilon_\tau), \qquad \epsilon_\tau \sim p(\epsilon_\tau), \qquad (\theta_l, \eta_l) \sim p_l(\theta_l, \eta_l).$$

Fourth, in latent-variable language, `tau` is not directly observed. It is a latent relational variable whose only allowed appearance in this paper is through a predefined image-channel projection. This is why compressed time-delay-distance posteriors are insufficient: the latent projection test requires access to image labels, parities, Fermat differences, and model uncertainties upstream of the final cosmographic posterior.

This bridge also clarifies what the paper does not claim. It does not claim a new mass component, a direct detection of a new field, or a replacement for standard time-delay cosmography. It defines a disciplined perturbation term and asks whether future evidence-grade lensing products can falsify it.

4. Why Lensing Is the Right Test Bed

The tau-core observer principle is an attempt to formalize a simple intuition: an observer does not see a gravitational source directly; the observer sees a projection of source geometry through accessible measurement channels. In ordinary weak-field applications this distinction can be hidden inside an effective potential. Strong lensing makes the distinction sharper. A lensed quasar, for example, is observed through multiple images of one source. These images share the same lens but differ in parity, path geometry, Fermat potential, arrival time, and local image constraints.

This makes time-delay lensing a natural test bed. If the observer principle has a lensing signature, it should not appear as an arbitrary per-lens fudge factor. It should appear as a shared, geometrically structured amplitude that uses image-level information and survives nuisance controls.

The central question of this paper is whether the tau-core observer principle can be turned into a falsifiable time-delay-lensing test, and what data products are required to run it.

Our answer is yes at the protocol level, but not yet at the real-data proof level. We define the test criterion, test its identifiability synthetically, audit public product availability, document the data products actually used, and close a static-control branch.

5. The Observer Principle in a Lensing System

For a time-delay lens, a standard lens model gives image positions, time-delay relations, and Fermat-potential differences. Let image `i` have a parity indicator and a Fermat-potential difference relative to a reference image. The tau-core observer principle predicts that an admissible observer-channel correction should depend on the multi-image geometry, not merely on a scalar distance posterior.

We therefore define the candidate T2 channel:

`T2_parity_abs_fermat`

The name is intentionally operational:

- `parity`: the term is sensitive to image parity or a model-derived parity proxy;
- `abs_fermat`: the term uses absolute Fermat-potential structure rather than delay-only compression;
- `T2`: the term is the only synthetic branch promoted as an active tau-core candidate in this work.

The amplitude multiplying this channel is:

`epsilon_tau`

A crucial constraint is that `epsilon_tau` must be shared. If every lens, image, or exposure receives its own free amplitude, the term is not an observer principle; it is ordinary nuisance absorption.

6. Operator Origin and Model-Comparison Requirement

The current paper does not derive `T2_parity_abs_fermat` from a dynamical principle. It motivates it as the simplest candidate operator that uses three features special to time-delay lensing:

Ingredient	Motivation	Remaining weakness
image parity	distinguishes minima, saddles, and maxima of the arrival-time surface	no symmetry has yet been shown to force parity weighting
absolute Fermat structure	keeps sensitivity to image-level arrival-time geometry rather than only compressed delays	no variational derivation yet selects `|Delta phi|`
shared scalar amplitude	prevents the term from becoming per-lens residual fitting	no completed theory yet predicts a universal scalar coefficient

Thus T2 is a first candidate in an operator family, not a final theoretical necessity. A future dynamical tau-core theory would have to supply at least one of the following:

1. a symmetry argument that selects parity-weighted Fermat structure;
2. a weak-field expansion of a covariant action whose leading lensing correction reduces to T2;
3. a conservation or reciprocity relation that forbids nearby operators;

4. a hierarchical evidence comparison showing that T2 is favored over a predefined operator family after nuisance freedom is included.

For the present paper, the conservative statement is:

`T2 is a symmetry-motivated candidate operator, not a derived operator.`

The necessary future comparison set should include at minimum:

Operator	Role
$O_i^{(0)} = 0$	standard no-tau baseline
$O_i^{(1)} = \Delta\phi_i$	ordinary signed Fermat perturbation
$O_i^{(2)} = \pi_i \lvert\Delta\phi_i\rvert$	current T2 candidate
$O_i^{(3)} = \lvert\Delta\phi_i\rvert$	parity-free absolute-Fermat control
$O_i^{(4)} = \pi_i$	parity-only control
$O_i^{(5)} = L_i$	geometric path-length proxy
scrambled $O_i^{(2)}$	label/parity/Fermat falsification control

Only an evidence comparison against this family can answer the reviewer question "why not another operator?"

7. Conditional Weak-Field Derivation Sketch

The paper can go one step beyond naming T2 as a candidate. This section gives a conditional derivation route. It is not yet a theorem of a complete tau-core theory, but it states the geometrical assumptions under which the T2 structure is the leading allowed image-channel correction.

In the thin-lens weak-field limit, image positions are stationary points of the Fermat potential

$$\phi(\theta;\beta) = \frac{1}{2}|\theta-\beta|^2 - \psi(\theta), \qquad \nabla_\theta\phi = 0.$$

For image i, define the Hessian

$$H_i = \nabla_\theta\nabla_\theta\phi(\theta_i;\beta), \qquad \pi_i = \operatorname{sgn}\det H_i.$$

The sign π_i is the Morse parity of the stationary point. Minima and maxima have positive parity, while saddle images have negative parity. Thus parity is not an additional arbitrary label; it is a discrete invariant of the local arrival-time geometry for a fixed lens model.

Now assume that a first-order observer-channel correction obeys four minimal conditions:

1. **Stationary-image locality:** at leading order, the correction is evaluated on each stationary image branch.
2. **Reference-shift invariance:** adding a constant to the Fermat potential cannot change the observable correction, so only differences $\Delta\phi_i = \phi_i - \phi_{\rm ref}$ may enter.
3. **Tau-window dilation covariance:** the magnitude part of the correction is generated by a projection-window integral with $w(\tau) = \alpha/(1+\tau)$ and $T_i = |\Delta\phi_i|/\phi_\star$.
4. **Arrival-time variational origin:** the same correction can be represented as a branch term in an effective arrival-time functional.
5. **Path-reversal evenness:** the magnitude of the relational delay mismatch is invariant under source-observer reversal, so the leading scalar dependence is even in $\Delta\phi_i$.
6. **Morse-orientation sensitivity:** the correction changes orientation across saddle versus non-saddle branches, so the only local discrete sign available at leading order is π_i.

Under these assumptions, the lowest-order scalar image-channel correction has the form

$$\delta\Phi_i = \lambda\, \pi_i\, F(|\Delta\phi_i|) + \mathcal{O}(\lambda^2).$$

If the expansion is analytic in the small dimensionless Fermat offset after normalization by the lens time-delay scale, then

$$F(|\Delta\phi_i|) = c_1|\Delta\phi_i| + c_2|\Delta\phi_i|^2 + \cdots.$$

Keeping only the leading term gives

$$\delta\Phi_i = \epsilon_\tau\, \pi_i|\Delta\phi_i| + \mathcal{O}(\epsilon_\tau|\Delta\phi_i|^2, \epsilon_\tau^2),$$

which is precisely the T2 operator used in this paper.

This is the first concrete physics bridge in the manuscript. It says that T2 is not selected by numerology alone: it is the leading term if the correction is local on stationary images, invariant under Fermat zero-point shifts, dilation-covariant in the tau-core projection window, representable as an arrival-time functional correction, even under path reversal in magnitude, and sensitive to Morse orientation. The open problem is to derive these assumptions from a covariant action, reciprocity principle, or observer-construction rule rather than imposing them.

8. Minimal Theorem Attempt

This section states the strongest theorem-like claim currently available. It is not a theorem of nature; it is a conditional uniqueness result inside the phenomenological lensing construction.

Definitions. Let ϕ_i be the Fermat potential evaluated on image branch i, let $\Delta\phi_i = \phi_i - \phi_{\mathrm{ref}}$, and let

$$\pi_i = \operatorname{sgn} \det H_i$$

be the Morse parity of the image branch. Let an admissible tau-core branch correction be a scalar map

$$\mathcal{Q}_i = \mathcal{Q}(\Delta\phi_i, \pi_i; \epsilon_\tau)$$

added perturbatively to the arrival-time/Fermat observable.

Axioms.

1. **Branch locality.** To first order, $\mathcal{Q}_i$ depends only on the stationary branch i, not on an independently fitted per-lens amplitude.
2. **Fermat zero-point invariance.** $\mathcal{Q}_i$ is invariant under $\phi_i \mapsto \phi_i + C$, so it can depend on $\Delta\phi_i$ but not on the absolute zero of ϕ.
3. **Tau projection depth.** The magnitude dependence enters through a dimensionless projection window $T_i = |\Delta\phi_i|/\phi_\star$.
4. **Dilation-covariant tau weight.** The projection kernel is the regularized dilation-covariant density $w(\tau) = \alpha/(1+\tau)$.
5. **Path-reversal evenness.** The magnitude of the correction is invariant under $\Delta\phi_i \mapsto -\Delta\phi_i$.
6. **Morse orientation.** The only allowed first-order branch sign is the local Morse parity π_i.
7. **Analytic small-window limit.** The correction admits an expansion for $T_i \ll 1$.

Lemma 1: magnitude term. Axioms 3, 4, and 7 imply

$$\mathcal{P}(T_i) = \alpha \ln(1 + T_i) = \alpha T_i + \mathcal{O}(T_i^2).$$

Thus the leading magnitude term is proportional to $|\Delta\phi_i|$.

Lemma 2: sign term. Axioms 5 and 6 imply that the leading branch sign is not $\operatorname{sgn}(\Delta\phi_i)$, because the magnitude is even under path reversal. The remaining local discrete orientation supplied by the arrival-time surface is π_i.

Lemma 3: first-order uniqueness. Under the above axioms, any admissible first-order scalar correction has the form

$$\mathcal{Q}_i = C\,\epsilon_\tau\,\pi_i|\Delta\phi_i| + \mathcal{O}(\epsilon_\tau|\Delta\phi_i|^2, \epsilon_\tau^2),$$

where C is a convention-dependent normalization absorbed into ϵ_τ.

Conditional theorem. Within this phenomenological tau-lensing construction, the T2 operator is the unique leading first-order branch correction satisfying branch locality, Fermat zero-point invariance, tau-window dilation covariance, path-reversal evenness, Morse orientation, and small-window analyticity.

The proof is conditional because the axioms are not yet derived from a covariant action. The theorem attempt therefore upgrades T2 from a hand-chosen operator to the leading term of a specified tau-core lensing construction, but it does not yet prove that nature must realize that construction.

Proof gaps. A complete proof would still need to derive branch locality, path-reversal evenness, and Morse-orientation coupling from a deeper source-observer reciprocity principle or covariant action. It would also need to show how the modified stationarity condition propagates into image positions, magnifications, and time-delay posteriors without introducing uncontrolled gauge artifacts.

9. Axiom Origin and Covariant Closure

The conditional theorem clarifies the next physics problem: the issue is no longer the algebraic form of T2, but the origin of the axioms. At present the axioms have different strengths.

Axiom	Current origin	Status
Fermat zero-point invariance	standard lensing convention; only time-delay differences are observable	physically well motivated
Morse orientation	standard stationary-point geometry of the Fermat surface	geometrically well motivated
tau-window dilation covariance	inherited tau-core projection symmetry	tau-core motivated, not covariantly derived
analytic small-window limit	ordinary effective-field/perturbative assumption	acceptable but regime-limited
branch locality	perturbative image-branch assumption	plausible, not derived
path-reversal evenness	candidate source-observer reciprocity principle	open

Axiom	Current origin	Status
Morse-orientation coupling	identification of parity as the branch orientation sign	plausible, but needs a reciprocity or variational theorem

This table is the current physics map. Some ingredients are standard lensing geometry; some come from the tau-core projection kernel; some are still imposed because the covariant completion is absent.

The missing covariant closure would need to define at least four objects:

1. **Primitive field or relational object.** A variable replacing the purely phenomenological $\tau(x;u)$, together with its transformation law.
2. **Invariant observable.** A scalar or relational quantity that reduces to the tau projection depth T_i in the thin-lens limit.
3. **Conserved or constrained current.** A relation showing why the tau correction cannot be chosen independently on each image branch.
4. **Reciprocity law.** A source-observer or path-reversal statement explaining why the magnitude is even while the branch orientation is carried by π_i.

A schematic covariant closure would therefore have the form

$$S[g,\psi,\tau,u] \quad \Longrightarrow \quad \nabla_\mu J^\mu_\tau = 0 \quad \Longrightarrow \quad \delta\Phi_i = \epsilon_\tau \pi_i |\Delta\phi_i| + \cdots$$

in the weak-field thin-lens limit. The present paper supplies only the last arrow conditionally. The first two arrows are the open theoretical task.

This distinction matters because the current theorem attempt could fail in two ways. It could fail empirically if the operator family comparison disfavors T2. Or it could fail theoretically if a covariant completion selects a different reciprocity law, a different projection depth, or a different branch orientation structure.

10. Candidate Reciprocity Principle

The most important missing physical ingredient is a reciprocity principle. The candidate principle proposed here is deliberately narrow: it is a condition on the observer-channel correction in the thin-lens weak-field limit, not yet a full covariant theorem.

Let R denote source-observer reversal or, equivalently at this level, orientation reversal of the image branch in the arrival-time surface. An admissible tau-core branch correction is required to decompose into a scalar projection depth and an orientation factor:

$$\delta\Phi_i = \epsilon_\tau \, \Omega_i \, \mathcal{P}(T_i),$$

with the transformation rules

$$R : T_i \mapsto T_i, \qquad R : \mathcal{P}(T_i) \mapsto \mathcal{P}(T_i), \qquad R : \Omega_i \mapsto -\Omega_i.$$

The first two conditions state why the magnitude is even: the projection depth is a recoverability or accessibility magnitude, not an oriented delay. In the lensing reduction this is

$$T_i = \frac{|\Delta\phi_i|}{\phi_\star}.$$

The third condition states that orientation must be carried by a local orientation pseudoscalar of the image branch. For a two-dimensional Fermat surface, the canonical local candidate is the Morse orientation

$$\Omega_i = \pi_i = \operatorname{sgn}\det H_i.$$

Thus the reciprocity candidate gives

$$\delta\Phi_i = \epsilon_\tau\, \pi_i\, \mathcal{P}\left(\frac{|\Delta\phi_i|}{\phi_\star}\right).$$

Using the tau-core kernel and expanding for small T_i gives T2 at leading order. This explains, conditionally, why the magnitude is even and why parity carries orientation.

The same principle also suggests why the amplitude should be shared. If ϵ_τ is the coupling of the projection current rather than a per-image nuisance parameter, then different image branches sample the same underlying tau-accessibility flux. The covariant version would require a current or constraint such as

$$\nabla_\mu J^\mu_\tau = 0 \quad \text{or} \quad \mathcal{C}_\tau[g, \tau, u] = 0,$$

whose thin-lens reduction leaves one shared coefficient rather than independent branch amplitudes.

This is still a candidate principle. The missing theorem is:

$$\text{covariant reciprocity} \quad \Longrightarrow \quad R : T_i \mapsto T_i, \quad R : \Omega_i \mapsto -\Omega_i, \quad \Omega_i = \pi_i.$$

Until that theorem exists, path-reversal evenness, Morse-orientation coupling, and shared amplitude are physically motivated but not derived.

11. Reciprocity Theorem To Be Proven

The central theory task can now be stated more sharply. The desired theorem is not "T2 fits data." It is a covariant statement explaining why the thin-lens observer-channel correction must decompose into an even projection magnitude and an oriented Morse factor.

Candidate theorem statement. Let $\mathcal{O}_\tau[g, \tau, u; \gamma]$ be the detector-accessible tau-core observable associated with a null path γ, where g is the metric, τ is the relational projection object, and u denotes the observer construction. Suppose there exists a source-observer reversal map R such that:

$$\mathcal{O}_\tau[R\gamma] = \sigma(\gamma)\, \mathcal{O}_\tau[\gamma],$$

where $\sigma(\gamma)$ is an orientation character of the null branch. The theorem needed by this program is:

$$\sigma(\gamma_i) \longrightarrow \pi_i = \operatorname{sgn}\det H_i, \qquad |\mathcal{O}_\tau[\gamma_i]| \longrightarrow \mathcal{P}\left(\frac{|\Delta\phi_i|}{\phi_\star}\right)$$

in the weak-field thin-lens limit.

If this theorem were proven, then the leading branch correction would follow:

$$\delta\Phi_i = \epsilon_\tau \pi_i \mathcal{P}\left(\frac{|\Delta\phi_i|}{\phi_\star}\right) = \epsilon_\tau \pi_i |\Delta\phi_i| + \cdots.$$

The theorem requires a covariant measurable. The paper currently has only candidates:

Candidate measurable	Thin-lens target	Problem
relational delay functional along γ	$\lvert\Delta\phi_i\rvert$	needs invariant definition under observer/source exchange
projection-depth scalar $T[\gamma]$	$\lvert\Delta\phi_i\rvert/\phi_\star$	normalization and gauge dependence must be controlled
tau current flux through an observer screen	shared ϵ_τ	no current J^μ_τ is derived
branch-orientation character $\sigma(\gamma)$	π_i	must be tied to Morse parity, not inserted by hand

This theorem can fail in precise ways. If no covariant scalar reduces to $|\Delta\phi_i|$, the magnitude part fails. If the reversal character is not Morse parity, the parity weighting fails. If no conserved or constrained tau current exists, the shared amplitude fails. If all three fail, T2 remains only a phenomenological operator.

Thus the real theoretical milestone is:

$$\boxed{\text{covariant measurable} + \text{reciprocity map} + \text{orientation character} \Longrightarrow \pi_i|\Delta\phi_i|}$$

The present paper reaches the right-hand side conditionally. It does not yet derive the boxed implication.

12. Theory Companion Summary and Open Gates

The preceding sections contain the theory needed inside this paper: the observer-channel correction is treated as a conditional, reciprocity-motivated weak-field candidate, not as a completed prediction of the Tau Core parent program. The longer derivation program now lives in the Tau Core theory hub, where it can be developed without overloading the empirical feasibility paper.

At paper level, the relevant conditional structure is:

$$\begin{aligned} &\text{screen-gauge invariance} + \text{ reciprocity-even arrival magnitude} \\ &\quad + \text{ branch locality} + \text{ shared observer-channel amplitude} \\ &\Longrightarrow \quad \delta\Phi_i = \epsilon_\tau \Sigma_i \mathcal{P}(T_i^{\mathrm{cov}}) + O(\epsilon_\tau^2). \end{aligned}$$

In the weak-field thin-lens limit,

$$\Sigma_i \to \pi_i, \qquad T_i^{\mathrm{cov}} \to \frac{|\Delta\phi_i|}{\phi_\star},$$

so the candidate leading term becomes

$$\delta\Phi_i = \epsilon_\tau \pi_i \mathcal{P}\left(\frac{|\Delta\phi_i|}{\phi_\star}\right) = \epsilon_\tau \pi_i |\Delta\phi_i| + O(\epsilon_\tau |\Delta\phi_i|^2, \epsilon_\tau^2).$$

The detailed theory companion is the Tau Core theory-hub lensing module. It tracks the longer proof route and the open theory gates:

Gate	Paper status	Theory-hub role
full reciprocity theorem	conditional theorem target	prove or refute the orientation/magnitude decomposition

Gate	Paper status	Theory-hub role
covariant measurable	candidate via Jacobi orientation and arrival-time scalar	derive the detector-accessible invariant
tau-accessibility current	shared-amplitude candidate	derive conservation or a constrained observer-channel law
screen interaction	controlled effective insertion	derive from boundary variation or replace with a better coupling
dynamical closure / GR limit	requirements stated	derive stress-energy, conservation, and decoupling from a parent action

This paper therefore uses the theory only to justify a falsifiable operator family and a data protocol. It does not claim that the parent theory has already forced the T2 operator uniquely.

13. Missing Geometric Necessity

The strongest remaining physics gap is not statistical. It is geometric. The paper defines an observable framework, a candidate operator family, a conditional weak-field route, and a future inference procedure, but it does not yet show that the T2 structure is forced by a deeper geometry.

A stronger theory would need at least one geometrical selection mechanism:

Missing mechanism	What it would need to show
symmetry necessity	a transformation principle that singles out parity-weighted Fermat structure
variational derivation	an action whose weak-field lensing limit produces the T2 operator
reciprocity theorem	a source-observer or path-reversal identity that requires the same correction
caustic/parity theorem	a result tying saddle/minimum parity structure to the residual sign or weight
dynamical selection rule	a reason the shared scalar amplitude is universal or hierarchical

At present, none of these has been derived. This means the physics kernel is embryonic: the paper has a disciplined way to test a candidate geometrical idea, but it does not yet contain the theorem that would make the candidate inevitable.

The most accurate classification of the work is therefore:

```
new physics claim: not established
geometrical necessity: not derived
phenomenological test kernel: defined
falsification and inference protocol: strong
```

This distinction is important. A negative future test would falsify the proposed T2 phenomenology, not every possible tau-core theory. A positive future test would motivate a deeper geometrical derivation, but would not by itself complete the theory.

14. Breakthrough Roadmap

The current manuscript is not the breakthrough. It is the scaffold that states what a breakthrough would have to look like. Three next results would change the scientific status of the program.

The primary theoretical game changer would be a weak-field derivation. If the T2 structure emerged as the leading correction in a controlled weak-field expansion,

$$\Phi_i = \Phi_i^{\mathrm{GR}} + \lambda \pi_i |\Delta\phi_i| + \mathcal{O}(\lambda^2),$$

then the program would move from a phenomenological operator test to a candidate geometrical theory. The crucial point is not merely the appearance of $|\Delta\phi_i|$, but the joint emergence of parity weighting, absolute Fermat structure, and a shared amplitude.

The most attractive geometrical route to such a result may be a reciprocity or parity theorem. A theorem of this kind would connect source-observer exchange, path reversal, or caustic crossing structure to the sign/weight carried by the image parity π_i. The target statement would have the schematic form:

$$\mathcal{G}_{\mathrm{lens}} \quad \Longrightarrow \quad \delta\Phi_i \propto \pi_i |\Delta\phi_i| \quad \text{at leading order.}$$

The first serious empirical milestone would be real-data operator competition. The decisive comparison is not T2 versus nothing in isolation, but T2 against the operator family $\{O^{(0)}, \dots, O^{(5)}\}$ after marginalizing lens-model nuisance freedom:

$$Z_2 > Z_m \quad \text{for relevant controls } m \neq 2,$$

with stability under prior sensitivity and posterior-predictive checks.

Until one of these results exists, the paper should be read as a disciplined phenomenological and falsification framework, not as a completed physics theory.

15. Likelihood and Evidence Framework

Because the paper uses population-level language, the future real-data test must be stated as a likelihood problem. For lens `l` and image-level data vector `d_l`, let `theta_l` denote ordinary lens parameters and `eta_l` denote nuisance/systematic parameters. For an operator `O_m`, the model is:

$$d_l \sim p\big(d_l \mid \theta_l, \eta_l, \epsilon_m, O^{(m)}\big)\,, \qquad \epsilon_m \sim p(\epsilon_m), \qquad (\theta_l, \eta_l) \sim p(\theta_l, \eta_l).$$

The posterior for a candidate operator is:

$$p\big(\epsilon_m, \{\theta_l\}, \{\eta_l\} \mid \{d_l\}, O^{(m)}\big) \propto p(\epsilon_m) \prod_l p(\theta_l, \eta_l) \prod_l p\big(d_l \mid \theta_l, \eta_l, \epsilon_m, O^{(m)}\big)\,.$$

The evidence for the operator is:

$$Z_m = \int p(\epsilon_m) \prod_l p(\theta_l, \eta_l) \prod_l p\big(d_l \mid \theta_l, \eta_l, \epsilon_m, O^{(m)}\big)\; d\epsilon_m \, d\theta_l \, d\eta_l.$$

The relevant comparison is not simply whether `epsilon_m` is nonzero. It is whether the T2 operator improves predictive and evidential performance after the Occam penalty for the shared coefficient and after marginalizing ordinary lens-model freedom. The no-tau comparison is:

$$B_{20} = \frac{Z_2}{Z_0}.$$

The operator-competition requirement is to compare Z_m across the predefined operator family. Posterior-predictive simulations must reproduce the residual structure without creating false positives in controls, and the conclusion must be stable under reasonable priors on ϵ_m and nuisance widths.

The present paper does not compute these real-data evidences because the required image/model products are not available in the public-only route. It does, however, define the likelihood objects that a future evidence-grade test must instantiate.

16. What Would Count as a Future Proof?

We use the word proof operationally, not metaphysically. A gravitational-lensing proof of the observer principle would require all of the following:

1. A public or auditable image/model product chain for one or more time-delay lenses.
2. A no-T2 model that reproduces a published Ddt or null posterior within a preregistered tolerance.
3. Image labels, positions, parity or parity-computable models, relative delays, and Fermat-potential differences.
4. A T2 perturbation with one shared `epsilon_tau`, not per-object amplitudes.
5. A likelihood-level posterior for `epsilon_tau`, with nuisance parameters marginalized.
6. Evidence or predictive improvement over the no-tau baseline and over nearby operator controls.
7. Prior-sensitivity and posterior-predictive checks.
8. A null result when the T2 channel is removed or scrambled.

The proof condition can be summarized as:

In words, a future proof would require a successful standard no-T2 reproduction, a statistically identified shared `epsilon_tau`, and survival under nuisance, scrambling, and false-positive controls.

This paper satisfies the last three items only in synthetic/control form. It does not satisfy the first two real-data requirements.

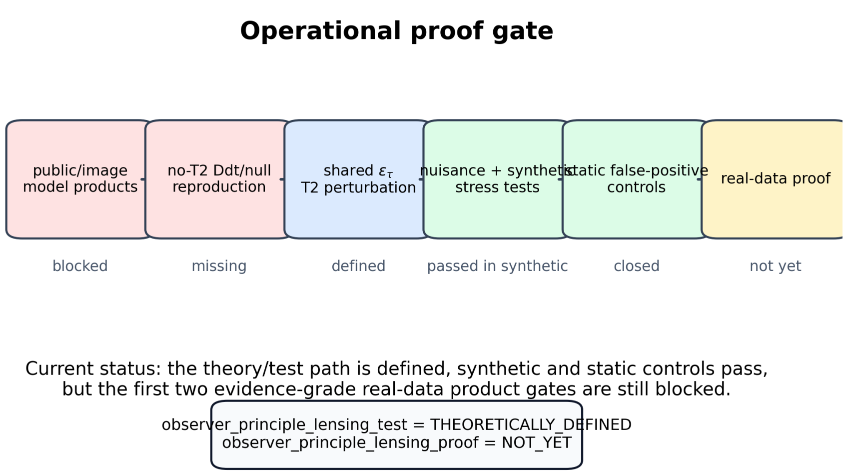

Figure 3: Figure 3. Operational proof gate. The T2 perturbation is not allowed until an evidence-grade no-T2 Ddt/null reproduction is available. Current work defines the test and validates synthetic/static controls, while the first two real-data product gates remain blocked.

17. Why Distance Posteriors Alone Are Insufficient

Published time-delay distance products are valuable cosmography products, but they are too compressed for the observer-gated T2 proof. A Ddt posterior alone does not expose which image parity, Fermat-potential difference, and image-position constraints produced it. Since the T2 channel is image-geometry dependent, the proof must begin upstream of compressed distance chains.

The required product is therefore not merely:

```
published Ddt posterior
```

but rather:

```
image/model products -> no-T2 Ddt reproduction -> controlled T2 perturbation
```

This distinction is the main practical reason the real-data branch is blocked.

18. Synthetic Identifiability Result

The synthetic branch asks whether the proposed T2 channel is mathematically identifiable before any real-data claim is attempted. The promoted synthetic term was:

```
T2_parity_abs_fermat
```

It was promoted only after numerical recovery tests and nearby synthetic controls indicated that it was distinguishable from simpler Fermat-like alternatives.

Nearby controls did not receive the same status:

Term	Role	Status
`T1_fermat`	ordinary Fermat-like control	expected control fail
`T2`: parity-weighted absolute Fermat	active tau-core branch	promoted in synthetic tests
`T3`: path-weighted proxy	proxy branch	not promoted as physics claim

The main hierarchical synthetic result was:

```
lens_count = 28
position_constraint_rows = 224
injected epsilon_tau = 0.05
epsilon_hat = 0.0499193089
epsilon_error = -0.0000806911
```

Additional synthetic stress results:

```
blind_challenge_scenarios = 4
strict_identified_count = 4
max_abs_epsilon_error = 0.0003890250
```

Noise Monte Carlo:

```
replicates_per_noise = 64
stable_noise_max = 0.008
transition_noise_min = 0.010
fail_noise_min = null
```

Joint stress grid:

```
scenario_count = 8
replicates_per_scenario = 48
stable_count = 6
transition_count = 2
fail_count = 0
```

Interpretation: the synthetic recovery is a generative sanity check, not evidence for new physics. Recovering an injected $\epsilon_\tau = 0.05$ mainly shows that the estimator can recover the signal class it was asked to recover under controlled assumptions. Its value is methodological: it checks identifiability, code behavior, stress degradation, and nearby-control behavior before any real-data claim is attempted. It does not show that nature contains the T2 operator.

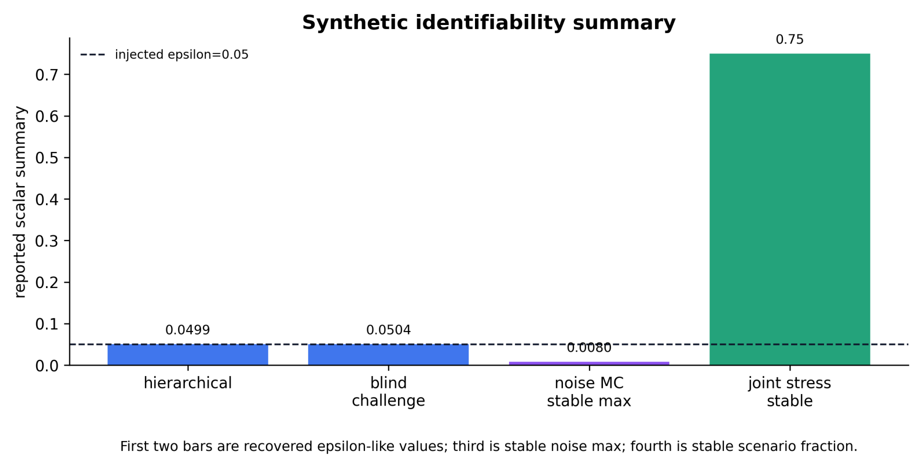

Figure 4: Figure 4. Synthetic identifiability summary. The recovered synthetic epsilon-like values remain near the injected amplitude, while cached stress tests map where stability begins to degrade.

19. Public Time-Delay Product Audit

The public audit asked whether the required proof gate can be run without private communication or unpublished products. The answer was no: the reviewed public products did not provide the target-specific posterior/configuration and Fermat/parity releases needed for this test.

The blocker is:

```
public products are compressed, paper-only, or software-only rather than
target-specific posterior/configuration and Fermat/parity releases
```

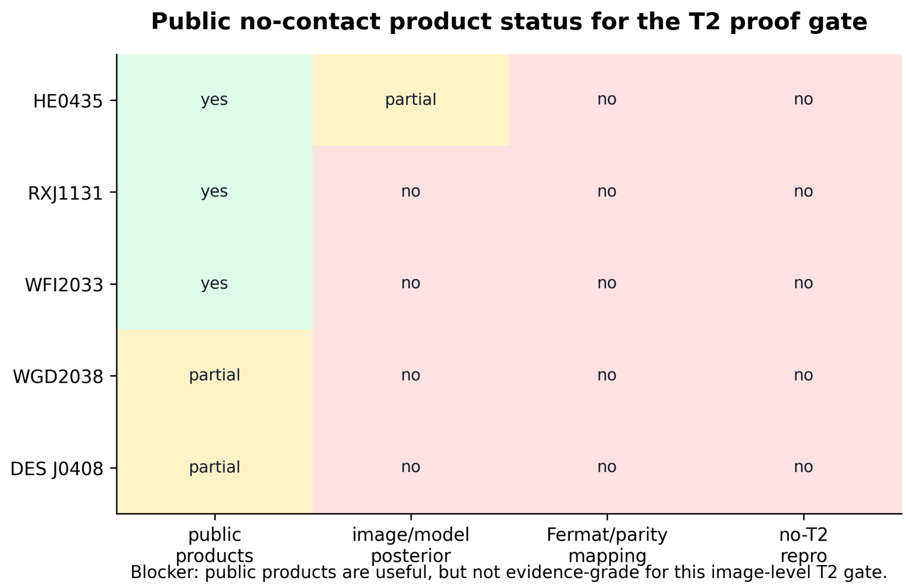

Figure 5: Figure 5. Public no-contact product status for the T2 proof gate. The reviewed targets have useful public products, but not the target-specific image/model posterior and Fermat/parity mapping needed for the no-T2 reproduction gate.

19.1 HE0435-1223

HE0435-1223 was the most developed public fallback. Public HST imaging can be ingested, and the reproduction stack defined Ddt tolerance, label conventions, pixel masks, null-model requirements, PSF policy, and archive protocol.

The extracted Ddt tolerance was:

```
q16/q50/q84 = 2539.56384848 / 2707.4290739999997 / 2889.194643
median tolerance = 54.14858148
```

However, the public-only model-level PSF correction attempt did not reach an evidence-grade reconstruction:

```
selected_hard_bound_solution_count = 6
baseline_grid_ring_median_rms = 14.4715078608
selected_ring_median_rms = 14.4505423335
improvement_vs_baseline = 0.0209655273
max_ring_five_sigma_fraction = 0.2173664667
```

This is not a criticism of the published HE0435 analysis. It says only that our public-only reconstruction is not evidence-grade enough to support a tau-core perturbation.

19.2 WGD2038-4008

WGD2038-4008 became the most concrete public follow-up route after the TDCOSMO2025 and target-repository audit. The public support payload contains Ddt samples, Ddt weights, kappa/environment support, and notebook-level hooks for Fermat-potential comparisons. We therefore ran a bounded HST-to-lenstronomy reproduction pass.

We also acquired the WGD2038 arXiv source-table payload from the official arXiv 2406.02683 source bundle. The materialized tables contain the published $D_{\Delta t}$ and flat-ΛCDM H_0 summary constraints, including the final $D_{\Delta t} = 1.68^{+0.40}_{-0.38}$ Gpc and $H_0 = 65^{+23}_{-14}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ entries. This is a source-backed published summary payload, not the missing posterior/Fermat payload: it contains no image-wise Fermat differences, sample identifiers, parity/order, or model posterior rows.

The result is a partial positive and a continued blocker. Public HST products were reduced into the three-band lenstronomy input format:

```
data_f160w.hdf5, psf_f160w.hdf5, psf_f160w_hires.hdf5
data_f475x.hdf5, psf_f475x.hdf5
data_f814w.hdf5, psf_f814w.hdf5
```

The multiband notebook data/setup path then passed a local preflight. One reproducibility detail was also identified: in the public F160W preprocessing notebook, the standard data and PSF save cells are raw cells, so a documented compatibility copy was used to execute only those unchanged save cells.

Finally, a bounded local model-plumbing smoke run was executed from the reconstructed three-band inputs. The smoke copy ran a PEMD/SHEAR configuration with one tiny PSO step and no PSF iteration, image alignment, MCMC, or posterior analysis. Because the local environment did not provide a buildable `fastell4py` backend, the smoke copy used `suppress_fastell=True`; this is a dependency bypass for model-plumbing only and is not a physical PEMD posterior.

After repairing the local build toolchain and installing `fastell4py` from the public source tree, the same bounded one-step PEMD/SHEAR smoke route was rerun without `suppress_fastell`. This verifies that the physical PEMD backend is available for local model-plumbing. It still does not constitute posterior reproduction, convergence, or physical validation.

We also added a WGD2038 holdout extraction contract for the DES-frozen one-amplitude score. The contract does not run a score and does not sample a T2 posterior; it specifies the required independent-holdout table before such a test would be admissible. The required fields are target/model/sample identifiers, image labels and ordering, parity or Morse type, image coordinates, $\Delta\phi_{AB}$, $\Delta\phi_{AC}$, $\Delta\phi_{AD}$, observed-delay or $D_{\Delta t}$ linkage, weights, and the no-T2 residual vector. The public Fermat-potential notebook exposes 36 expected joblib/posterior targets, but the posterior outputs and image-wise Fermat/parity/arrival table are not materialized in the current public package. Thus the WGD2038 route is now contract-defined, not score-complete.

The WGD2038 partial holdout materialization then materialized the public part of this contract into a 36-row model-level manifest. This records the expected WGD2038 model targets together with the public $D_{\Delta t}$, kappa/environment, redshift, velocity-dispersion, and lens-property support. The processed public payload contains 567880 $D_{\Delta t}$ samples, with weighted median $D_{\Delta t} \approx 1493.69$. This is a source-backed partial materialization, not a score table: zero rows currently contain the per-sample image labels, parity/order, Fermat differences, and no-T2 residual vector needed to apply the DES-frozen score.

The WGD2038 bounded local Fermat preflight then tested whether the local, explicitly non-converged diagnostic outputs can at least support the image-level extraction machinery. Three bounded local jobs expose four-image best-fit tables, giving 12 image rows with image coordinates, Fermat potentials, Jacobian determinant/trace, and Morse/parity labels. For the primary profile-freeze v2 diagnostic, the notebook-basis Fermat differences are approximately $\Delta\phi_{AB} = -0.0060$, $\Delta\phi_{AC} = -0.0294$, and $\Delta\phi_{AD} = -0.0809$. This is a technical extraction preflight only: it does not use a published or converged WGD posterior, does not produce a no-T2 residual vector, and still leaves zero score-ready rows.

The WGD2038 observed-delay linkage audit then closes the next question after the Fermat preflight. The TDCOSMO XVI publication reports new WGD2038 time-delay measurements, uses delays relative to image A, and uses the corresponding covariance matrix in the cosmographic inference. We therefore added a WGD2038 observed-delay no-T2 smoke artifact that transcribes the Fig. 2 delay vector and covariance into a local machine-readable record. In the publication convention $\Delta t_{AX} = t_A - t_X$, the transcribed delays are $AB = -12.4$, $AC = -5.3$, and $AD = -33.3$ days, with the covariance matrix recorded in the AB, AC, AD order. The artifact also records the sign-convention bridge from the WGD notebook basis, where $\Delta\phi_{AB}$ denotes $\phi_B - \phi_A$, to the A-centered publication basis. Combining this vector with the local bounded Fermat preflight gives a no-T2 residual smoke. This is not a WGD holdout score: the Fermat input is a local explicitly non-converged diagnostic, not a published or converged WGD posterior. The remaining blocker is therefore narrower but still decisive: a score-ready WGD Fermat/posterior table in the same image convention is still needed before the DES-frozen score can be applied.

We also added a WGD2038 published-model delay-shape crosscheck. This uses the TDCOSMO IX combined GLEE and lenstronomy no-T2 delay predictions at $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, together with the TDCOSMO XVI observed-delay vector and covariance. Fitting a single global delay scale to the published predictions gives the best scalar match for the lenstronomy combined prediction, with scale 1.1015, an implied pure-scale $H_0 \simeq 63.55\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and a remaining normalized delay-shape residual of approximately 2.29 sigma units against the observed-delay covariance. This is a published-model-level crosscheck, not a posterior-level WGD score: the model-prediction uncertainties and posterior covariance are not fully propagated because the score-ready posterior/Fermat table is still missing. No T2 parameter is fitted or sampled.

We then freeze a WGD2038 delay-shape holdout target from this crosscheck. In the A-centered AB, AC, AD delay basis, the predeclared target residual is $(-6.8925,\ 5.7151,\ -6.6435)$ days, with covariance-metric norm 3.2378 sigma units. Future WGD posterior/Fermat products can be compared against this fixed direction using the recorded covariance-metric cosine, rather than choosing a direction after seeing the posterior-level result. This target is deliberately conservative: it is not endpoint-blind, because it is defined from the WGD2038 observed-delay vector and published model predictions; it is not a posterior-level score; and it is not T2 evidence.

The next bounded pilot added a deliberately tiny `emcee` path after one PSO step. This verifies that the local fastell-backed posterior-plumbing route can execute and write an output artifact. It is not a converged posterior, evidence-grade no-T2 reproduction, or authorization for T2 sampling.

A short diagnostic pilot then used one PSO step followed by a 20-step `emcee` path. In the lenstronomy environment this produced a finite $(4240, 53)$ sample payload with finite log-probabilities. A simple split-half mean-shift check gave median and maximum shifts of approximately 0.093 and 0.426 standard deviations, respectively. This is a bounded chain-health diagnostic only, not a convergence certificate.

A longer bounded diagnostic used one PSO step followed by a 60-step `emcee` path. The derived diagnostic summary records a finite $(12720, 53)$ sample payload with finite log-probabilities. The split-half mean-shift quantiles are approximately 0.377 standard deviations at the median, 0.833 at the 90th percentile, and 0.943 at the maximum. This is stronger evidence that the local posterior route is operational, but the visible drift means it is still not a converged posterior.

A 120-step diagnostic produced a finite $(25440, 53)$ sample payload. However, the split-half mean-shift quantiles increased to approximately 0.820 standard deviations at the median, 1.398 at the 90th percentile, and 1.493 at the maximum. We treat this as a useful negative convergence diagnostic: the physical MCMC route runs, but simple lengthening from this initialization does not yet produce a stable no-T2 posterior.

A diagnostic120 continuation run was then started from the previous chain endpoint rather than from a fresh PSO initialization. It again produced a finite $(25440, 53)$ payload with finite log-probabilities. The split-half mean-shift quantiles were approximately 0.817 standard deviations at the median, 1.261 at the 90th percentile, and 1.412 at the maximum. This confirms that the continuation path is operational, but it is still a bounded chain-health diagnostic rather than a converged no-T2 posterior.

A second continuation from that endpoint also completed and produced a finite $(25440, 53)$ payload. Its split-half mean-shift quantiles were approximately 0.873 standard deviations at the median, 1.375 at the 90th percentile, and 1.526 at the maximum. This is a useful negative diagnostic: repeated endpoint continuation alone does not remove the no-T2 posterior convergence blocker.

Finally, a cold continuation from the second endpoint lowered the MCMC proposal scale to $\sigma_{\text{scale}} = 0.02$. It also produced a finite $(25440, 53)$ payload, but the split-half mean-shift quantiles worsened to approximately 0.983 standard deviations at the median, 1.409 at the 90th percentile, and 1.519 at the maximum. Thus simple proposal-scale reduction does not clear the convergence blocker either.

We therefore inspected the parameter-level split-half drift across the bounded diagnostic120 and continuation payloads. The largest persistent drifts are concentrated in nuisance-heavy lens/source-light profile and image-position directions, including indexed image positions and Sersic radius/index parameters in the lens and source light models. This diagnoses the remaining reproduction blocker more sharply: the local route is operational, but the nuisance posterior has not stabilized. The derived nuisance-stabilization plan therefore selects a bounded `profile_freeze_v1` follow-up: stabilize the highest-drift lens/source light-profile nuisance directions first, keep image positions under audit, and keep mass/shear and T2 claims protected. That follow-up was then run as a bounded diagnostic. Since fixing nuisance parameters changes the active parameter set, it used previous best-fit initialization but did not reuse old MCMC samples. It produced finite $(20160, 42)$ samples with finite log-probabilities. The split-half mean-shift quantiles improved to approximately 0.633 standard deviations at the median, 1.206 at the 90th percentile, and 1.422 at the maximum. This is a positive nuisance-stabilization diagnostic, but it is not a converged no-T2 posterior. An aggressive `profile_freeze_v2` follow-up then fixed additional Chameleon and light-profile nuisance directions while still leaving image positions under audit and mass/shear unfrozen. It remained finite with $(16800, 35)$ samples, but the split-half mean-shift quantiles worsened to approximately 1.038 standard deviations at the median, 1.454 at the 90th percentile, and 1.516 at the maximum. We therefore treat v2 as a useful negative diagnostic: after v1, simply freezing more light-profile directions is not the next clean route.

This improves the practical status of the WGD2038 route, but it does not satisfy the no-T2 proof gate. The original cluster/MCMC joblib outputs, or an equivalent bounded local physical posterior reproduction with convergence diagnostics, are still missing. Therefore no WGD2038 real-data T2 sampling is authorized.

19.3 Other Targets

RXJ1131-1231 and WFI2033-4723 were found to have useful public material, but the available products were too compressed for the image-level T2 reproduction required here.

After the WGD2038 nuisance-stability blocker was isolated, we performed an alternate-source audit rather than continuing to freeze nuisance directions. This changes the status of DES J0408-5354. The public `DESJ0408_time_delay_cosmography` repository contains five lenstronomy-oriented notebooks, six processed data/PSF HDF5 files, 24 lens-model posterior files, 24 time-delay posterior files, and a velocity-dispersion grid. DES J0408 is promoted to bounded no-T2 follow-up, not to evidence. The next required step is a compatibility extractor and a bounded no-T2 baseline reproduction from those public products. No DES J0408 real-data T2 posterior was sampled, and no T2 sampling is authorized by this audit.

We then ran the first DES J0408 no-T2 time-delay extraction smoke. This reads the 24 public time-delay posterior files directly, without adding any T2 parameter, and compares their two delay columns to the observed delay constants declared by the public post-processing class:

```
observed_delays = [-112.1, -155.5] days
observed_sigma  = [   2.1,   12.8] days
model_count = 24
sample_count_per_model = 10000
```

The best unweighted smoke diagnostic is the run917 composite model, whose full identifier is recorded in the derived audit table. Its mean chi-square is approximately 3.657 against the two observed delays, and 0.9928 of its samples fall inside the two-dimensional 2-sigma box. This is a useful compatibility result: it shows that DES J0408 contains a directly readable no-T2 time-delay posterior layer. It is not yet an evidence-grade no-T2 reproduction because the full lens posterior, model weights/logZ, and image/model quantities have not yet been decoded or rerun locally.

We therefore added a DES J0408 full posterior compatibility smoke. The public posterior files require only import-compatibility patches for older Astropy and dill serialization. With those patches, all 24 lens-model posterior files load successfully. Each contains 10,000 samples, 57–62 active parameters, a matched public time-delay posterior file, point-source image-position records, logZ-like sampler fields, and model-configuration metadata. The highest logZ-like record in this smoke is a composite model, while the best unweighted time-delay diagnostic above is also in the composite family. This strengthens DES J0408 as the next no-T2 reproduction target, but it still stops before the proof gate: image-level Fermat or arrival-time quantities have not yet been recomputed from the decoded posterior samples.

The next DES J0408 arrival-time recomputation smoke was deliberately bounded to a compatible power-law model. It successfully turns decoded posterior samples into finite image-level arrival-time differences under the current lenstronomy reader. However, those recomputed differences do not reproduce the public time-delay table for the same model. For the first 128 samples, the mean absolute discrepancies are about 87 and 136 days in the two delay columns. We therefore first treated this as a negative compatibility result, not as physics evidence. A pair audit also shows that simple image-pair relabeling does not resolve the raw-reader mismatch: the best direct pair choice still has an RMSE of about 90 days.

The same audit then identifies the compatibility recovery. The old posterior stores 57 parameters, while modern lenstronomy expects 58 because it inserts `s_scale_lens0` after `gamma_lens0`; duplicate `ra_image` and `dec_image` entries must also be matched by occurrence rather than by a plain dictionary key. With occurrence-aware old-to-modern alignment, `tau0 -> tau0_list` aliasing, and `s_scale_lens0=0`, the recomputed arrival-time differences reproduce the public DES time-delay table with RMSE about 0.0145 days on the first 128 samples. This is still not a T2 result, but it materially improves the DES J0408 route: the immediate blocker becomes promotion of this alignment to a reusable extractor across the compatible model family.

The DES J0408 power-law family alignment smoke applies that recovery to all 12 public power-law posterior files. It confirms that the raw modern reader is not reliable for these legacy files. The 57-parameter core subset is largely recovered: 5 of 6 models pass the strict public-table match criterion, with median aligned RMSE about 0.017 days. The 60-parameter legacy subset remains blocked: 0 of 6 models pass under the current rule, with median aligned RMSE about 8.22 days. This is not yet a complete no-T2 baseline. It is, however, a sharper operational split: the 57-parameter power-law path can now support a bounded feature-table extraction, while the 60-parameter and composite paths require separate legacy compatibility handling.

We then promoted that split to a DES J0408 57-parameter core feature-table audit. This longer-prefix check recomputes 1,024 arrival-time samples for the validated core models and compares them to the public time-delay tables without introducing T2. It further narrows the usable baseline: only 2 of the 5 previously recovered core models remain clean under both row-wise and distributional feature checks. The best model by public observed-delay chi-square and by core-relative logZ weight is `0408_run1001_0_0_0_0_0_1_1_0`, which reproduces the public prefix with RMSE about 0.015 days. This creates a strict DES no-T2 feature artifact, but it also preserves the negative result: the full 57-parameter core is not yet a completed no-T2 baseline.

A follow-up failure diagnostic shows that this narrowing is not caused by a broad ordinary-row mismatch. The four failed 57-parameter models are outlier-dominated under the current alignment: most rows remain close to the public table, but one or two catastrophic legacy rows drive the strict RMSE failure. Without an independently justified row-level provenance or outlier policy, those models are not promoted. The strict DES no-T2 feature core therefore remains the two clean models only.

A row-level provenance audit further refines this blocker. For the top catastrophic rows in the four failed models, the posterior coordinates are not parameter outliers, the log-likelihoods are not in the lowest one percent of the inspected prefix, and no runtime warnings are emitted. The rows therefore look like ordinary posterior samples whose public/recomputed table pairing breaks. This supports a row-linkage/provenance interpretation of the compatibility blocker, but it does not justify row removal or promotion of the failed models without an independent linkage rule from the public notebooks or serialized sampler records.

We then audited the public source for such a rule. The public `output_class.py` supports intended index-order linkage: `compute_model_time_delays()` loops over `samples_mcmc[i]`, computes Δt_{AB} and Δt_{AD}, and appends the corresponding row; `load_time_delays()` later loads the saved table without reordering and asserts that its length matches the chain. The distance posterior notebook loads the public `td_` tables through this path. This narrows the interpretation of the blocker, but it still does not provide an independent outlier-removal or row-recovery policy. Consequently the failed 57-parameter models remain unpromoted unless the original legacy lenstronomy row semantics can be reconstructed.

The runtime evidence supports this as a plausible compatibility issue. The public distance-posterior notebook states that it runs with `lenstronomy v0.9.2`, with notebook metadata indicating a Python `2.7.15` kernel. The helper runtime used here is Python `3.9.6` with `lenstronomy 1.14.1`, Astropy `6.0.1`, and dill `0.4.1`. This version gap is a plausible source of the few row-level legacy defects, but it is not treated as proof: no legacy environment has been reconstructed, and no failed model is promoted on this basis.

We also ran an optional local `lenstronomy 0.9.2` compatibility probe under Python `3.9.6`. The probe uses NumPy `1.23.5`, SciPy `1.9.3`, Astropy `5.0.8`, and a small SciPy private-API shim needed by the old reader. This reconstructed reader imports and runs, but it does not recover the public DES J0408 time-delay rows: all six 57-parameter core models fail the rowwise check over the first 128 samples, with the best RMSE still approximately 622 days. This does not refute the original Python `2.7.15` production environment, but it is negative evidence against using this Python-`3.9` `lenstronomy 0.9.2` workaround to promote the four failed 57-parameter models. The strict DES no-T2 baseline therefore remains the two clean 57-parameter models.

Finally, we converted the two strict clean DES rows into a lensing-feature to Tau-role constraint artifact. This is not a physical $\mathrm{Response}_\tau^{\mathrm{lens}}$ law. It is a cautious data-side narrowing of the common morphology candidate. The clean DES rows force six weak roles in the lensing role cover: endpoint-blind provenance, internal mid/mass geometry, line-of-sight/environment structure, source-readout anchoring, rowwise closure stability, and observed-delay scale compatibility. The same artifact preserves the negative constraint that the failed DES models cannot be promoted without an independent row-recovery or null-policy proof. Thus the lensing branch now constrains the shared morphology candidate, while the full Tau-side lensing response, amplitude calibration, and real-data T2 claim remain unproved.

We also added a DES J0408 no-T2 time-residual candidate pretest. This returns to the original observer-gated question, but in a deliberately bounded form: before fitting any T2 parameter, ask whether the strict clean no-T2 models leave a coherent residual direction in the observed time delays. For the two clean 57-parameter DES rows, the unweighted model-minus-observed residual vector is approximately

$$\Delta t_{\mathrm{model}} - \Delta t_{\mathrm{obs}} \simeq (-3.43,\ +19.84)\ \mathrm{days},$$

while the logZ-weighted vector is approximately $(-2.81, +19.95)$ days. The signs agree across both clean models, and the pairwise residual-vector cosine is about 0.998. This is not a Tau-specific time-shift detection: it does not fit or sample T2, it excludes the failed DES rows, and it may still reflect ordinary lens-family mismatch, source modeling, time-delay likelihood structure, or other systematics. Its role is narrower but useful: it supplies a specific no-T2 residual direction for a future bounded T2 design, while leaving real-data T2 sampling unauthorized.

As a negative control, the same audit also evaluates the non-clean DES feature rows rather than silently discarding them. These rows show a similar residual direction. Therefore residual-direction coherence alone is not specific enough to support a T2 or Tau-time claim. The scientifically safe interpretation is that DES J0408 now provides a concrete pre-registered design vector for a later null-versus-T2 comparison, not a detected observer-centered time distortion.

We therefore freeze a DES J0408 null-versus-T2 design freeze artifact. The frozen correction direction is the observed-minus-model vector,

$$\Delta t_{\mathrm{obs}} - \Delta t_{\mathrm{model}} \simeq (+3.43,\ -19.84)\ \mathrm{days},$$

in the two-delay basis. The same artifact freezes the active null competitors: ordinary no-T2 nuisance freedom, lens/source-family mismatch, DES row/runtime systematics, and the non-clean-row coherence control. This does not authorize a real-data T2 fit. It only prevents a future comparison from choosing its target direction after seeing the result.

With this design frozen, we can define the one-amplitude T2 operator pretest

$$\Delta t_{\mathrm{corr}} = \Delta t_{\mathrm{noT2}} + \alpha u_{\mathrm{frozen}},$$

where u_{frozen} is the pre-frozen unit direction and $\alpha \geq 0$ is chosen by closed-form least squares on the two strict clean DES rows. This reduces the clean-core residual RMSE from about 20.15 days to about 0.62 days. Simple orthogonal, swapped-component, and opposite direction controls do not improve the score under the same nonnegative-amplitude rule. However, the same frozen amplitude also improves the non-clean DES control rows. Therefore this is a useful directional score pretest, not a T2 discovery: the operator is endpoint-derived, not yet Tau-derived, and not evidence of a physical observer-centered time distortion.

Finally, we add an independent holdout readiness audit for this DES-frozen score. The result is negative but useful. WGD2038-4008 is the best current holdout route, with readiness score 3/6, but it is not score-ready: the extracted image-wise Fermat/arrival table, image parity/order, original joblib posterior payload, and a converged no-T2 posterior are still missing. Therefore the DES one-amplitude score remains design-only. The next finite action is to extract or reconstruct the WGD2038 per-sample image/model table with image labels, parity/order, $d\phi_{AB}/d\phi_{AC}/d\phi_{AD}$, model sample IDs, and observed-delay or Ddt linkage.

These targets remain scientifically valuable. They are blocked only for this specific observer-gated proof gate because the required image/model product depth was either too compressed or, in the DES J0408 case, not yet converted into an evidence-grade no-T2 reproduction.

20. Static-Lens Controls

Because real time-delay proof products were unavailable, we developed a static control analysis to test false-positive behavior. The branch used HFF cluster morphology controls: Chandra gas, HFF lensing mass maps, and public member-catalog tracers.

The analysis completed five control layers:

Control layer	Role
HFF morphology selection	select an independent static-lens morphology layer
control scorecard	require heterogeneous controls

Control layer	Role
null injection	test zero-T2 false positives
null stress sweep	bracket the noise/threshold boundary
systematic-template sweep	flag dangerous templates

The scorecard was heterogeneous:

```
strong_control = 1
mixed_control = 2
weak_negative_control = 1
selected_static_control = macs0717
selected_negative_control = abells1063
```

Baseline null injection:

```
injected_epsilon_tau = 0.0
observed_null_epsilon_hat = 0.004678323836468474
false_positive_rate = 0.0
replicates = 2048
```

Stress boundary:

```
reference_threshold = 0.05
reference_threshold_stable_max_noise = 0.020
first transition at threshold 0.05: noise_sigma = 0.030
```

Template sensitivity:

```
nominal_templates_all_stable = true
inverted_morphology: transition, false_positive_rate = 0.010498046875
residual_aligned_adversarial: fails, false_positive_rate = 0.70947265625
```

The static controls support the falsification discipline of the proof program: the workflow does not automatically see tau-core structure everywhere. But static controls contain no time-delay posterior and therefore cannot prove the observer principle.

21. Anticipated Reviewer Objections

The most likely reviewer objections are valid, and the paper should be read with them in view.

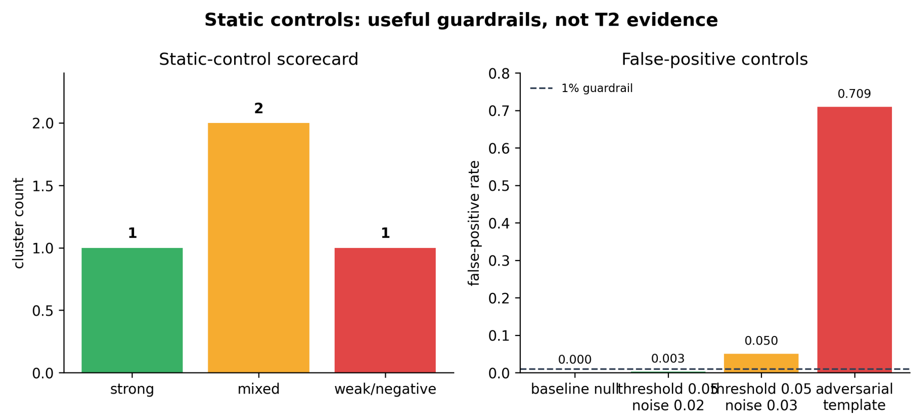

Figure 6: Figure 6. Static-control guardrails. HFF morphology controls remain heterogeneous, pass the baseline null-injection test, and expose noise/template regimes where false positives can be manufactured.

Objection	Correct response in this paper
The T2 operator looks handcrafted.	Partly. The manuscript now gives a conditional weak-field route, but T2 is still not a derived necessity until the assumptions are obtained from an action, reciprocity theorem, or covariant observer construction.
The synthetic recovery is too clean.	Yes. It is a sanity check showing recovery from a known generative class, not evidence that the class is realized in nature.
There is no real-data posterior.	Correct. No no-T2 Ddt/null reproduction, real-data Bayes factor, or posterior-predictive check is claimed.
The tau-core background theory is too open.	Correct. The paper does not yet specify what fundamental object transforms, what is invariant, or what the primitive measurable is in a complete theory.
The axioms are imposed, not derived.	Correct. The paper now maps their possible origins, but path-reversal evenness, Morse-orientation coupling, and covariant closure remain open theory tasks.

Objection	Correct response in this paper
The reciprocity principle is missing.	Partly. The thin-lens reciprocity lemma is proven, but no full covariant reciprocity theorem is claimed.
There is no covariant measurable.	Partly. The paper now proposes an optical measurable candidate using observer proper-time depth and Jacobi-map orientation, but it is not yet derived from a full tau-core action.
The shared amplitude is imposed.	Partly. The paper now gives a tau-accessibility current candidate, but no conservation theorem has yet been proven.
The current has no action origin.	Partly. The paper gives a minimal Noether-current route from tau shift symmetry, but no complete covariant action is claimed.
The screen construction may be gauge-dependent.	Partly. The paper states screen-gauge covariance conditions, but a full action-level derivation is still missing.
There is no dynamical closure.	Correct. The paper now states the required stress-energy, conservation, and GR-limit conditions, but does not derive them from a complete action.

These objections do not invalidate the narrower contribution. They define its scope. The present manuscript is best classified as a methods and feasibility note with a physics kernel, not as a new-physics detection paper.

22. Main Proposition of the Paper

We can now state the paper's central theorem in operational form.

Proposition. An operational gravitational-lensing test of the tau-core observer principle is theoretically possible because the observer principle can be mapped to an image-level time-delay lensing operator family with a shared population-level amplitude. The present candidate operator is `T2_parity_abs_fermat`. This proposition does not assert that a complete tau-core action, field content, covariance structure, conservation law, GR-limit derivation, gauge interpretation, or operator-selection principle has been supplied. A future empirical proof would require the T2 amplitude to be identified only after a no-T2 Ddt/null reproduction, likelihood-level nuisance marginalization, evidence comparison against nearby operators, and synthetic and static false-positive controls.

Status. The present work verifies the synthetic and control components of this theorem, but not the real-data no-T2 reproduction component.

Therefore:

```
tau-lensing perturbation test: defined at phenomenological Level 1
real-data proof: not yet achieved
complete tau-core field theory: not yet supplied
dynamical operator-selection principle: not yet supplied
geometrical necessity theorem: not yet supplied
missing ingredient: evidence-grade time-delay image/model products
```

This is the strongest defensible conclusion.

23. Data Used and Reproducibility

The present work used three categories of material.

First, public time-delay lensing product information was audited through local source manifests and product probes. This included HE0435-1223, RXJ1131-1231, WFI2033-4723, WGD2038-4008, DES J0408-5354, H0LiCOW public distance products, TDCOSMO public notebook routes, and related public repositories or metadata pages. These products were sufficient to define blockers but not sufficient to authorize the no-T2 reproduction gate.

Second, synthetic lensing data were used to test whether the proposed T2 channel is identifiable in principle. These tests do not use real lensing measurements as evidence; they test the mathematical behavior of the proposed estimator under controlled noise, nuisance, and position-constraint assumptions.

Third, static HFF cluster products were used as false-positive controls. These include already-generated summaries from Chandra gas, HFF lensing mass maps, and member-catalog morphology controls. They test whether the workflow becomes over-eager in systems without time-delay evidence.

Primary packet:

Primary packet:

```
studies/tau_core_lensing_observer_gate_v01/paper_packet_v01_public_blocked/
```

Critical scripts:

```
repro/scripts/check_repro.py
repro/scripts/build_pdfs.py
studies/tau_core_lensing_observer_gate_v01/make_manuscript_pdf.py
studies/tau_core_lensing_observer_gate_v01/make_arxiv_source.py
```

Critical outputs:

```
outputs/tau_core_lensing_observer_gate_v01/
repro/results/*/summary.json
```

Run:

```
python repro/scripts/check_repro.py
python studies/tau_core_lensing_observer_gate_v01/make_manuscript_pdf.py
python studies/tau_core_lensing_observer_gate_v01/make_arxiv_source.py
python -m pytest -q
```

Expected:

```
REPRO_CHECK_PASS
4 passed
```

24. Limitations

1. No real-data T2 posterior was sampled.
2. No public-only target passed the no-T2 Ddt/null reproduction gate.
3. The HE0435 public fallback failed PSF/model validation and is not a published-posterior reproduction.
4. The WGD2038 public HST route now passes a lenstronomy preprocessing and multiband setup preflight, and a bounded model-plumbing smoke run completes under documented compatibility constraints. A second bounded smoke run also completes with the physical `fastell4py` PEMD backend, and a minimal `emcee` posterior-plumbing pilot writes an output artifact. Short finite diagnostic pilots also complete, including a 60-step run with 12720 finite samples and a 120-step run with 25440 finite samples. The 120-step run shows increased split-half drift, so this is still not a converged physical cluster/MCMC posterior reproduction and does not authorize real-data T2 sampling.
5. Synthetic tests validate method behavior, not astrophysical truth.
6. Static controls are false-positive controls, not time-delay evidence.
7. The contact route for original products was disabled by user choice in this work session.
8. The tau-core observer principle is not yet a closed physical theory: action, field content, covariance structure, symmetry, conservation relation, first-principles GR limit, and gauge meaning remain open.
9. The T2 operator is not yet dynamically derived. The paper gives a conditional weak-field route, but its assumptions remain to be derived from a deeper theory and it must be tested against a predefined operator family.
10. No real-data likelihood, evidence, Bayes factor, prior-sensitivity analysis, or posterior-predictive check has yet been computed.
11. No geometrical necessity theorem, reciprocity theorem, variational derivation, or symmetry selection rule has yet been supplied for T2.
12. The paper has not yet defined the primitive physical measurable of `tau(x; u)` in a complete theory: what transforms, what remains invariant, and what detector-accessible quantity carries the tau-core correction.
13. The theorem attempt depends on axioms whose origins are only partially identified; path-reversal evenness, Morse-orientation coupling, conserved

tau current, and covariant closure remain open.

14. The reciprocity principle is proven only as a thin-lens lemma. A covariant theorem deriving the measurable, the reversal map, and the current structure has not yet been supplied.
15. No covariant detector-accessible tau-core measurable has yet been derived; the paper lists candidates but does not prove their invariance or thin-lens reduction.
16. The optical measurable $\mathcal{O}^{\mathrm{cov}}_{\tau,i} = \Sigma_i \mathcal{P}(T^{\mathrm{cov}}_i)$ is a candidate reduction, not yet a consequence of a full action or conserved tau current.
17. The tau-accessibility current J^{μ}_{τ} is only a closure candidate; no conservation theorem has yet been derived that forces a shared ϵ_{τ}.
18. The Noether-current route assumes a tau shift symmetry and a screen interaction term; neither has yet been derived from a complete covariant tau-core action.
19. Screen-gauge covariance is specified only as a consistency condition; the distinction between physical orientation reversal and arbitrary screen relabeling remains to be derived from a complete observer construction.
20. Dynamical closure is not derived: the stress-energy contribution, conservation relation, screen response, and GR limit remain requirements for a future complete theory.

25. Conclusion

Gravitational lensing provides a natural route toward testing the tau-core observer principle because a single lens supplies multiple observer-accessible image channels. This paper formulates the required proof criterion and identifies `T2_parity_abs_fermat` as the operative synthetic channel. The channel is identifiable in controlled synthetic tests and is protected by static false-positive controls.

The proof is not yet complete, and the underlying tau-core theory is not yet complete either. The physics kernel is embryonic: no geometrical necessity theorem, symmetry selection rule, reciprocity argument, or variational derivation currently forces T2. Public no-contact products also do not provide the evidence-grade image/model chain needed to reproduce a no-T2 Ddt/null posterior. The line is therefore scientifically alive at the phenomenological test level, but the real-data claim, the operator-necessity claim, and the full physical theory claim remain open.

Later Tau Core lensing work separates a complementary photon-lensing Morphological Alignment Residual route from the T2 time-delay route studied here. In that protocol, a source-side morphology template $T_{\tau} = B_{\mathrm{env}} A_f M_{bf}$ is compared with convergence/shear residuals under orientation, shuffle, baryon, filament, systematics, and no-Tau controls. This does not alter the present paper's claim boundary: the T2 observer-gated operator remains a phenomenological feasibility and falsification object, and MAR-type morphology correlations would still require shared-parent covariance controls before they could be read as more than lensing/morphology support.

The next decisive step is not another static control. It is an evidence-grade time-delay lens product chain that permits:

`no-T2 Ddt/null reproduction -> shared-epsilon T2 perturbation -> nuisance and false-positive controls`

Only that sequence can turn the proof program into an empirical proof.

References and Source Anchors

- Suyu et al. / H0LiCOW public data products: https://shsuyu.github.io/H0LiCOW/site/h0licow_data.html
- H0LiCOW Paper I overview: https://academic.oup.com/mnras/article/468/3/2590/3055701
- TDCOSMO I public notebook path: https://github.com/TDCOSMO/TD_data_public/blob/master/TDCOSMO_I/PSTD_notebook.ipynb
- TDCOSMO I time-delay cosmography methods PDF: https://lss.fnal.gov/archive/2019/pub/fermilab-pub-19-665-scd.pdf
- TDCOSMO XXVI Zenodo metadata: https://zenodo.org/records/19421439
- Local feasibility ledger: `feasibility_ledger.md`